

Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen				Direct Printing Legend	GBR
F,Mask		0,02mm		Solder Resist	Solder Mask GBR
Copper	L1 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L2 (GND)	0,0152mm (0,43oz)		Plane	GBR
Core		0,5579mm	FR4	Dielectric	
Copper	L3 (Sig, PWR)	0,0152mm (0,43oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L4 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Core		0,5579mm	FR4	Dielectric	
Copper	L5 (GND)	0,035mm (1,00oz)		Signal	GBR
Prepreg		0,1mm	FR4	Dielectric	
Copper	L6 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
B,Mask		0,02mm		Solder Resist	Solder Mask GBR
B,Silkscreen				Direct Printing Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,6262mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

aeris_pcb Fabrication Document

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY PALROC LABS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE102.500 x 90.000 mm

BOARD THICKNESS1.626 mm

TRACE WIDTH0.152 mm

TRACE TO TRACE0.150 mm

MIN. HOLE (PTH)0.150 mm

MIN. HOLE (NPTH)0.600 mm

ANNULAR RING0.100 mm

COPPER TO HOLE0.254 mm

COPPER TO EDGE0.250 mm

HOLE TO HOLE0.254 mm

Top Fabrication (Scale 1:1)

- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: Palroc Labs		Variant: PRELIMINARY	Git Hash: 604a55f
		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: palroc_one_pcb.kicad_pcb	Designer: Oriol Reverte	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 1 of 12